# VENKATA GURAVAREDDY

**PRINCIPAL PLATFORM ENGINEER | AI / GPU & BARE METAL INFRASTRUCTURE ARCHITECT**

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## EXECUTIVE SUMMARY

Principal Platform Infrastructure Engineer with **14 years** of foundational experience architecting, provisioning, and operating enterprise-scale distributed compute platforms. Deep subject matter expert spanning the full stack from bare metal hardware lifecycle (PXE, IPMI, BIOS, PCIe) to high-density GPU clusters (NVIDIA H100/L40S/DGX/HGX, InfiniBand, RDMA, NVLink) and Kubernetes platforms. Track record of scaling infrastructure from 2-person POCs to **42,000+ edge sites** and managing **600+ Kubernetes/OpenStack clusters** across 52 enterprise data centers. **Granted US Patent Holder (US11336525B1)** with 2 additional patents under review, and recipient of **20+ Verizon Spotlight Awards** for sustained technical excellence and high-impact platform delivery.

## CORE TECHNICAL COMPETENCIES

**AI & GPU Infrastructure:**
NVIDIA H100, A100, L40S, DGX/HGX, GPU Operator, DCGM, Fabric Manager, vLLM, LiteLLM, Triton Inference Server, CUDA, NVLink, InfiniBand, RoCE/RDMA.

**Networking & High Performance Fabric:**
TCP/IP, SR-IOV, DPDK, CNI (Calico/Cilium/Multus), BGP, VXLAN, Load Balancing (F5/Envoy), Cross-Cluster Routing, Multi-Site Mesh, TLS/mTLS.

**Platform & Cluster Engineering:**
Kubernetes (Core & Internals), OpenShift, OpenStack, KVM, Bare Metal Provisioning, IPMI/Redfish, PXE Boot, Kernel & BIOS Tuning, Linux (RHEL/Ubuntu).

**Storage, Automation & Telemetry:**
Ceph, MinIO, NetApp Trident (CSI), Python, Go, Bash, Terraform, Ansible, ArgoCD, Helm, GitOps, Prometheus, Grafana, ELK Stack, Jaeger/OpenTelemetry.

## PROFESSIONAL EXPERIENCE

### Verizon | Principal Engineer : AI & Cloud Infrastructure
**2017 - Present | Hyderabad, India**

**Lead Architect : Enterprise GPU Infrastructure & GenAI Inference Platform** — 2023 - Present

- **End-to-End GPU Platform Architecture:** Designed and deployed Verizon's centralized GPU inference compute platform from physical bare metal to model serving; provisioned and tuned multi-node NVIDIA H100 and L40S clusters running GPU Operator, DCGM telemetry, and NVIDIA Fabric Manager.
- **High-Throughput Model Serving:** Integrated and benchmarked low-latency LLM serving engines (vLLM, LiteLLM proxy, and Triton Inference Server) with custom KV-cache optimization, continuous batching, and dynamic multi-tenant GPU slicing, boosting inference throughput by 3.8x.
- **High-Speed Fabric & Low-Level Production Debugging:** Engineered high-bandwidth InfiniBand and RDMA fabric topologies; resolved critical production issues involving PCIe bus errors, XID anomalies, and NVLink inter-GPU saturation (caught and mitigated a critical 94% NVLink saturation incident using live DCGM profiling).
- **Platform Observability & P99 SLOs:** Instituted comprehensive Prometheus/Grafana GPU monitoring stacks tracking GPU SM utilization, memory thermal throttling, tensor pipe stalls, and P99 token generation latency, delivering 99.99% infrastructure uptime.

**Staff Platform Engineer : Fleet Kubernetes & Cloud Infrastructure** — 2021 - 2024

- **Large-Scale Multi-Site Fleet Operations:** Managed architecture and operations for **600+ Kubernetes and OpenStack production clusters** across 52 geographic data center sites, supporting thousands of business-critical microservices and 5G network functions.
- **Proprietary Capacity Framework (CER Engine):** Engineered "CER", Verizon's distributed capacity analytics and real-time reservation engine processing **5,000,000+ telemetry records/hour** across 500+ clusters and 60,000+ active pods; preempted cluster outages and reduced compute over-provisioning spend by $1.8M annually.
- **Bare Metal to Ingress Full Stack Lifecycle:** Automated firmware updates, IPMI power control, CNI overlay tuning, and persistent distributed storage lifecycle using Ceph and NetApp Trident CSI plugins.
- **Multi-Tenant Isolation & Security:** Enforced namespace isolation, pod disruption budgets, resource quotas, and admission control webhooks across shared multi-tenant clusters, eliminating cross-tenant interference.

**Platform Lead : Atlas 5G Cloud Native Network Platform** — 2017 - 2021

- **Architected & Scaled Atlas Platform:** Spearheaded the architectural design and implementation of Atlas, Verizon's enterprise-grade 5G Cloud-Native Network Function (CNF) deployment, validation, and admission control engine. Scaled the platform from an initial 2-person prototype to a distributed fleet operating across **42,000+ edge locations** and handling **10,000+ operations/hour**.
- **Patent Award & Vendor Disruption:** Awarded US Patent **US11336525B1** for automated multi-stage CNF pre-flight validation, container security profiling, and Kubernetes admission webhooks. Displaced legacy commercial solutions (HP NFVD), standardizing 5G service deployment corporate-wide.
- **Engineering Leadership:** Scaled and mentored the cross-functional engineering team from 2 to 14 engineers, governing architectural RFC reviews, CI/CD GitOps pipelines, code quality standards, and SRE operational practices.

## PREVIOUS CAREER EXPERIENCE

**Ericsson R&D** | **Senior Software Engineer** — **2016 - 2017 | Bangalore, India**

- **VNF Onboarding & SDN Controllers:** Built modular Python microservices and automated CI/CD validation harnesses for Virtualized Network Function (VNF) lifecycle orchestration integrated with OpenStack and SDN controllers.
- **Open Source Contribution:** Contributed upstream bug fixes and network virtualization automation code to open-source telecommunications and 5G working groups.
- **Infrastructure Verification:** Designed high-throughput network benchmarking pipelines verifying packet processing performance across virtual switches and compute hypervisors.

**Hewlett Packard Enterprise (HPE R&D)** | **Software Engineer** — **2014 - 2016 | Bangalore, India**

- **Bare Metal Automation:** Automated server provisioning, BIOS/firmware injection, and configuration management for HPE ProLiant Gen8/Gen9 rack servers and Moonshot systems using Python, IPMI, and OpenStack Cinder/Ironic.
- **On-Site International Deployment:** Selected for on-site US customer client deployment; led direct datacenter rack bring-up, SAN storage connectivity, and operational handoff for tier-1 enterprise clients.
- **Firmware & Storage Validation:** Validated kernel driver integrations and hardware RAID configurations to guarantee storage reliability under intensive I/O workloads.

**HCL Technologies (Cisco Practice)** | **Software Engineer** — **2012 - 2014 | Chennai, India**

- **Cisco Networking & TAC Collaboration:** Authored automated regression suites in Python for Cisco IOS and IOS-XE network operating systems; investigated and resolved escalated customer-reported production defects in partnership with Cisco TAC.
- **Hardware Lab Topology Simulation:** Designed and configured complex physical router/switch testbed topologies replicating customer multi-hop routing, VLAN trunking, and high-availability failovers.
- **Test Harness Engineering:** Built automated traffic generator harnesses to stress-test L2/L3 protocol stacks and isolate edge-case packet drop anomalies.

## KEY ARCHITECTURAL HIGHLIGHTS & QUANTIFIED SCALE

**42,000+ Edge Nodes**
Scaled the Atlas 5G engine nationwide, managing 10,000+ automated operations per hour with zero downtime.

**600+ K8s / OpenStack Clusters**
Architected centralized bare metal provisioning, telemetry, and storage across 52 distributed data center sites.

**3.8x GPU Inference Gain**
Optimized H100/L40S inference clusters using vLLM, continuous batching, and high-speed InfiniBand/RDMA fabric.

## PATENTS & INTELLECTUAL PROPERTY

GRANTED **US11336525B1:** *"Automated Containerized Network Function (CNF) Validation, Security Profiling, and Policy-Enforced Deployment System."* Invented core admission verification architecture protecting distributed 5G Kubernetes fleets.

PENDING **2 Patents Under USPTO Review:** Novel mechanisms for distributed AI/GPU workload telemetry arbitration and multi-cluster automated capacity balancing.

## HONORS & KEY RECOGNITIONS

★ **20+ Verizon Spotlight Awards**
Recognized across 8+ years for exceptional technical leadership, Sev-0/1 production incident resolution, GPU platform delivery, and cross-team excellence.

★ **Academic Gold Medalist**
Graduated Department Valedictorian with Highest Academic Distinction in Computer Science & Engineering (Andhra University, 2012).

## EDUCATION

**Bachelor of Technology (B.Tech) : Computer Science & Engineering**
Andhra University | 2012
Department Gold Medalist (First Rank across University)

## CERTIFICATIONS

✓ **Red Hat Certified Specialist in OpenShift Administration**
✓ **Cisco Certified Network Associate (CCNA)**